MTRC

MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE

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Branch: CSE , AI&DS,AI&ML

Batch: A,B ,AI&DS,AI&ML

Semester: Vth

Session: 2024-25(odd Sem.)

Unit No.: 3

Date of Submission: 05-10-2024

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References Book:

1. COMPUTER NETWORK,ANDREW S. TANENBAUM 5TH EDITION PEARSON PUBLICTION
2. COMPUTER NETWORKING JAMES F KUROSE PEARSON PUBLICATION
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FACULTY **HOD** **DEAN ACADEMICS**

UNIT 3

LECTURE 1

DESIGN ISSUES IN NETWORK LAYER

Network layer is majorly focused on getting packets from the source to the destination, routing error handling and congestion control.

Before learning about design issues in the network layer, let's learn about its various functions.

- Addressing**:**
  Maintains the address at the frame header of both source and destination and performs addressing to detect various devices in network.
- **Packeting:**
  This is performed by Internet Protocol. The network layer converts the packets from its upper layer.
- Routing**:**
  It is the most important functionality. The network layer chooses the most relevant and best path for the data transmission from source to destination.
- **Inter-networking:**
  It works to deliver a logical connection across multiple devices.

Network layer design issues:

The network layer comes with some design issues they are described as follows:

1. Store and Forward packet switching:

The host sends the packet to the nearest router. This packet is stored there until it has fully arrived once the link is fully processed by verifying the checksum then it is forwarded to the next router till it reaches the destination. This mechanism is called "Store and Forward packet switching."

2. Services provided to Transport Layer**:**

Through the network/transport layer interface, the network layer transfers it's services to the transport layer. These services are described below.

But before providing these services to the transfer layer following goals must be kept in mind:-

- Offering services must not depend on router technology.
- The transport layer needs to be protected from the type, number and topology of the available router.
- The network addresses for the transport layer should use uniform numbering pattern also at LAN and WAN connections.

Based on the connections there are 2 types of services provided:

- **Connectionless –** The routing and insertion of packets into subnet is done individually. No added setup is required.
- **Connection-Oriented –** Subnet must offer reliable service and all the packets must be transmitted over a single route.

3. Implementation of Connectionless Service**:**

Packet are termed as "datagram's" and corresponding subnet as "datagram subnets". When the message size that has to be transmitted is 4 times the size of the packet, then the network layer divides into 4 packets and transmits each packet to router via. a few protocol. Each data packet has destination address and is routed independently irrespective of the packets.

4. Implementation of Connection Oriented service:

To use a connection-oriented service, first we establishes a connection, use it and then release

it. In connection-oriented services, the data packets are delivered to the receiver in the same order in which they have been sent by the sender.
It can be done in either two ways:

- **Circuit Switched Connection –** A dedicated physical path or a circuit is established between the communicating nodes and then data stream is transferred.
- **Virtual Circuit Switched Connection –** The data stream is transferred over a packet switched network, in such a way that it seems to the user that there is a dedicated path from the sender to the receiver. A virtual path is established here. While, other connections may also be using the same path.

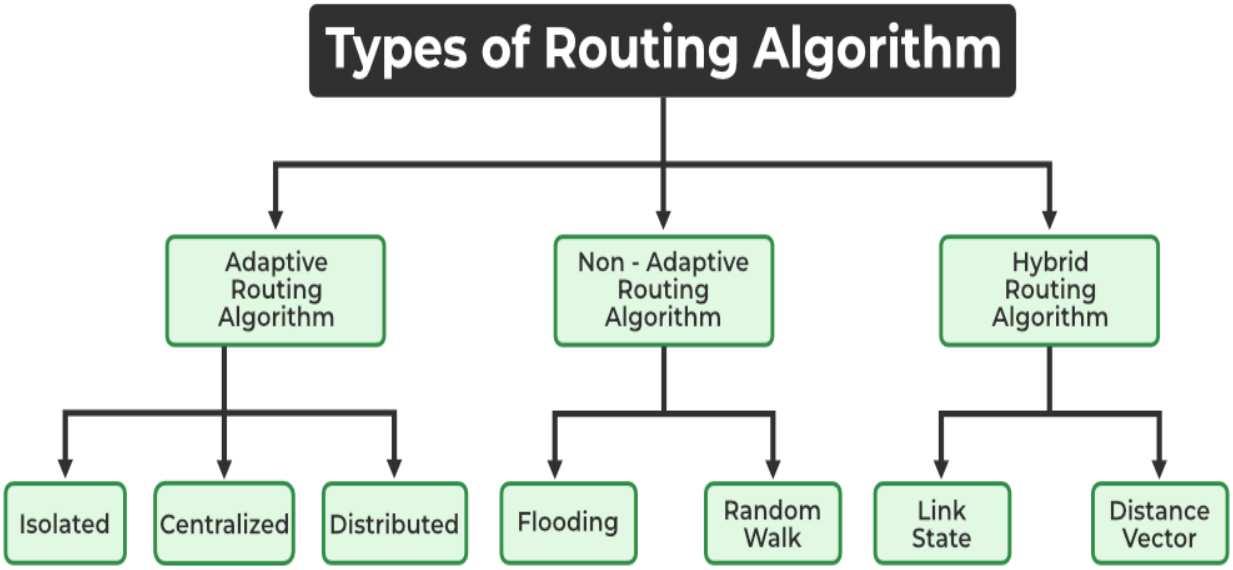
CLASSIFICATION OF ROUTING ALGORITHM

Routing is the process of establishing the routes that data packets must follow to reach the destination. In this process, a routing table is created which contains information regarding routes that data packets follow. Various routing algorithms are used for the purpose of deciding which route an incoming data packet needs to be transmitted on to reach the destination efficiently.

Classification of Routing Algorithms

The routing algorithms can be classified as follows:

1. Adaptive Algorithms
2. Non-Adaptive Algorithms
3. Hybrid Algorithms



1. Adaptive Algorithms

These are the algorithms that change their routing decisions whenever network topology or traffic load changes. The changes in routing decisions are reflected in the topology as well as the traffic of the network. Also known as dynamic routing, these make use of dynamic

information such as current topology, load, delay, etc. to select routes. Optimization parameters are distance, number of hops, and estimated transit time.
Further, these are classified as follows:

- **Isolated:** In this method each, node makes its routing decisions using the information it has without seeking information from other nodes. The sending nodes don't have information about the status of a particular link. The disadvantage is that packets may be sent through a congested network which may result in delay. Examples: Hot potato routing, and backward learning.

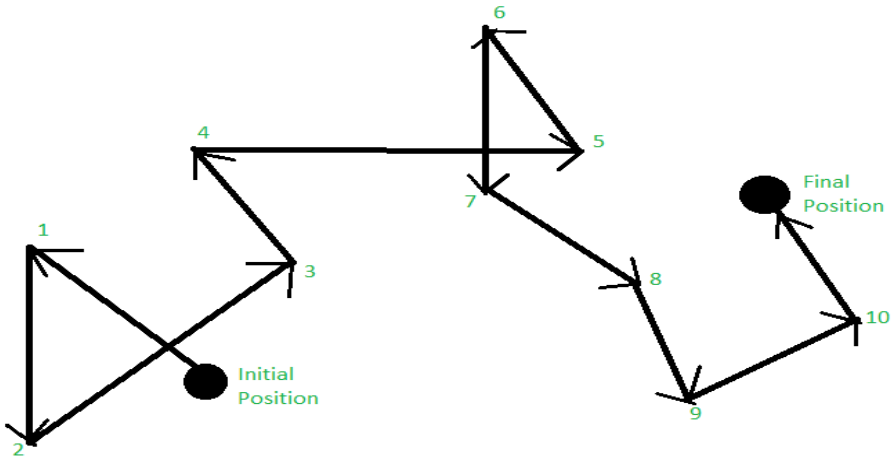
- **Centralized:** In this method, a centralized node has entire information about the network and makes all the routing decisions. The advantage of this is only one node is required to keep the information of the entire network and the disadvantage is that if the central node goes down the entire network is done. The link state algorithm is referred to as a centralized algorithm since it is aware of the cost of each link in the network.

- **Distributed:** In this method, the node receives information from its neighbors and then takes the decision about routing the packets. A disadvantage is that the packet may be delayed if there is a change in between intervals in which it receives information and sends packets. It is also known as a decentralized algorithm as it computes the least-cost path between source and destination.

2. Non-Adaptive Algorithms

These are the algorithms that do not change their routing decisions once they have been selected. This is also known as static routing as a route to be taken is computed in advance and downloaded to routers when a router is booted.
Further, these are classified as follows:

- **Flooding:** This adapts the technique in which every incoming packet is sent on every outgoing line except from which it arrived. One problem with this is that packets may go in a loop and as a result of which a node may receive duplicate packets. These problems can be overcome with the help of sequence numbers, hop count, and spanning trees.
- **Random walk:** In this method, packets are sent host by host or node by node to one of its neighbours randomly. This is a highly robust method that is usually implemented by sending packets onto the link which is least queued.



Random Walk

3. Hybrid Algorithms

As the name suggests, these algorithms are a combination of both adaptive and non-adaptive algorithms. In this approach, the network is divided into several regions, and each region uses a different algorithm.
Further, these are classified as follows:

- Link-state: In this method, each router creates a detailed and complete map of the network which is then shared with all other routers. This allows for more accurate and efficient routing decisions to be made.
- Distance vector: In this method, each router maintains a table that contains information about the distance and direction to every other node in the network. This table is then shared with other routers in the network. The disadvantage of this method is that it may lead to routing loops.

Difference between Adaptive and Non-Adaptive Routing Algorithms

The main difference between Adaptive and Non-Adaptive Algorithms is:

Adaptive Algorithms are the algorithms that change their routing decisions whenever network topology or traffic load changes. It is called Dynamic Routing. Adaptive Algorithm is used in a large amount of data, highly complex network, and rerouting of data.

Non-Adaptive Algorithms are algorithms that do not change their routing decisions once they have been selected. It is also called static Routing. Non-Adaptive Algorithm is used in case of a small amount of data and a less complex network.

For more differences, you can refer to Differences between Adaptive and Non-Adaptive Routing Algorithms.

Difference between Routing and Flooding

The difference between Routing and Flooding is listed below:

Routing	**Flooding**
A routing table is required.	No Routing table is required.
May give the shortest path.	Always gives the shortest path.
Less Reliable.	More Reliable.
Traffic is less.	Traffic is high.
No duplicate packets.	Duplicate packets are present.

LECTURE 2

SHORTEST PATH ROUTING

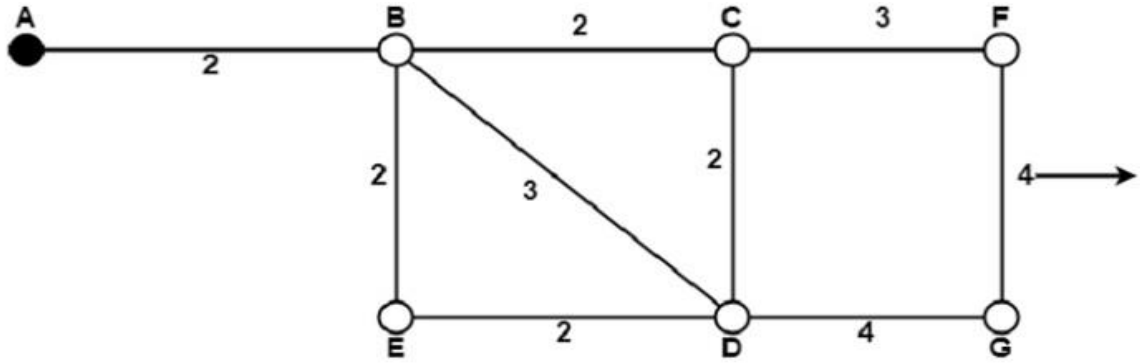
In this algorithm, to select a route, the algorithm discovers the shortest path between two nodes. It can use multiple hops, the geographical area in kilometres or labelling of arcs for measuring path length.

The labelling of arcs can be done with mean queuing, transmission delay for a standard test packet on an hourly basis, or computed as a function of bandwidth, average distance traffic, communication cost, mean queue length, measured delay or some other factors.

In shortest path routing, the topology communication network is defined using a directed weighted graph. The nodes in the graph define switching components and the directed arcs in the graph define communication connection between switching components. Each arc has a weight that defines the cost of sharing a packet between two nodes in a specific direction.

This cost is usually a positive value that can denote such factors as delay, throughput, error rate, financial costs, etc. A path between two nodes can go through various intermediary nodes and arcs. The goal of shortest path routing is to find a path between two nodes that has the lowest total cost, where the total cost of a path is the sum of arc costs in that path.

For example, Dijikstra uses the nodes labelling with its distance from the source node along the better-known route. Initially, all nodes are labelled with infinity, and as the algorithm proceeds, the label may change. The labelling graph is displayed in the figure.



Graphical Representation of Nodes with labeled path

It can be done in various passes as follows, with A as the source.

- Pass 1. B (2, A), C(∞,−), F(∞,−), e(∞,−), d(∞,−), G 60
- Pass 2. B (2, A), C(4, B), D(5, B), E(4, B), F(∞,−),G(∞,−)
- Pass 3. B(2, A), C(4, B), D(5, B), E(4, B), F(7, C), G(9, D)

We can see that there can be two paths between A and G. One follows through ABCFG and the other through ABDG. The first one has a path length of 11, while the second one has 9. Hence, the second one, as G (9, D), is selected. Similarly, Node D has also three paths from A

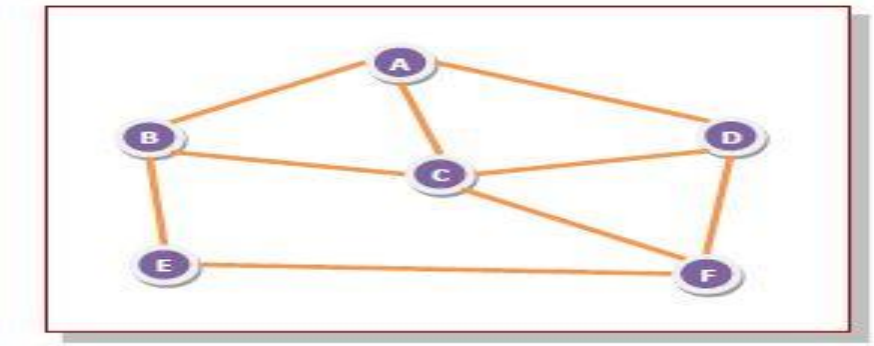
as ABD, ABCD and ABED. The first one has a path length of 5 rest two have 6. So, the first one is selected.

All nodes are searched in various passes, and finally, the routes with the shortest path lengths are made permanent, and the nodes of the path are used as a working node for the next round.

FLOODING

Flooding is a **non-adaptive routing technique** following this simple method: when a data packet arrives at a router, it is sent to all the outgoing links except the one it has arrived on.

For example, let us consider the **network** in the figure, having six routers that are connected through transmission lines.



Using flooding technique −

- An incoming packet to A, will be sent to B, C and D.
- B will send the packet to C and E.
- C will send the packet to B, D and F.
- D will send the packet to C and F.
- E will send the packet to F.
- F will send the packet to C and E.

Types of Flooding

Flooding may be of three types −

- **Uncontrolled flooding** − Here, each router unconditionally transmits the incoming data packets to all its neighbours.
- **Controlled flooding** − They use some methods to control the transmission of packets to the neighbouring nodes. The two popular algorithms for controlled flooding are Sequence Number Controlled Flooding (SNCF) and Reverse Path Forwarding (RPF).
- **Selective flooding** − Here, the routers don't transmit the incoming packets only along those paths which are heading towards approximately in the right direction, instead of every available paths.

Advantages of Flooding

- It is very simple to setup and implement, since a router may know only its neighbours.
- It is extremely robust. Even in case of malfunctioning of a large number routers, the packets find a way to reach the destination.
- All nodes which are directly or indirectly connected are visited. So, there are no chances for any node to be left out. This is a main criteria in case of broadcast messages.
- The shortest path is always chosen by flooding.

Limitations of Flooding

- Flooding tends to create an infinite number of duplicate data packets, unless some measures are adopted to damp packet generation.
- It is wasteful if a single destination needs the packet, since it delivers the data packet to all nodes irrespective of the destination.
- The network may be clogged with unwanted and duplicate data packets. This may hamper delivery of other data packets.

HIERARCHICAL ROUTING

In hierarchical routing, the **routers** are divided into regions. Each router has complete details about how to route packets to destinations within its own region. But it does not have any idea about the internal structure of other regions.

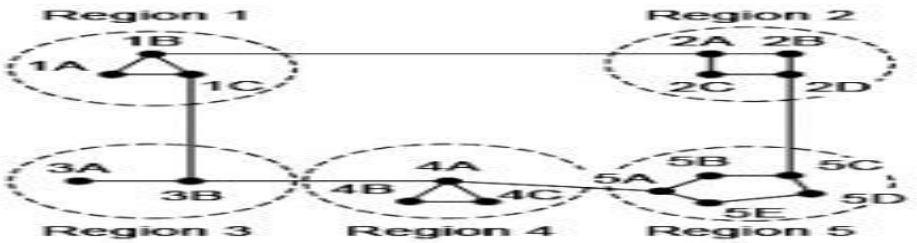
As we know, in both LS and DV algorithms, every router needs to save some information about other routers. When network size is growing, the number of routers in the network will increase. Therefore, the size of routing table increases, then routers cannot handle network traffic as efficiently. To overcome this problem we are using hierarchical routing.

In hierarchical routing, routers are classified in groups called regions. Each router has information about the routers in its own region and it has no information about routers in other regions. So, routers save one record in their table for every other region.

For huge networks, a two-level hierarchy may be insufficient hence, it may be necessary to group the regions into clusters, the clusters into zones, the zones into groups and so on.

Example

Consider an example of two-level hierarchy with five regions as shown in figure −



Let see the full routing table for router 1A which has 17 entries, as shown below −

Full Table for 1A

Dest.	Line	Hops
1A	-	-
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

When routing is done hierarchically then there will be only 7 entries as shown below −

Hierarchical Table for 1A

Dest.	Line	Hops
1A	-	-
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

Unfortunately, this reduction in table space comes with the increased path length.

Explanation

Step 1 − For example, the best path from 1A to 5C is via region 2, but hierarchical routing of all traffic to region 5 goes via region 3 as it is better for most of the other destinations of region 5.

Step 2 − Consider a subnet of 720 routers. If no hierarchy is used, each router will have 720 entries in its routing table.

Step 3 − Now if the subnet is partitioned into 24 regions of 30 routers each, then each router will require 30 local entries and 23 remote entries for a total of 53 entries.

Example

If the same subnet of 720 routers is partitioned into 8 clusters, each containing 9 regions and each region containing 10 routers. Then what will be the total number of table entries in each router.

LECTURE 3

Difference between Broadcast and Multicast

The process of delivering data from a source device to a destination device is known as transmission. In a network, data is converted into packets and sent to its destination. There are several methods of transmission, two of which are broadcast and multicast. In summary, multicasting is much more efficient and uses less bandwidth than broadcasting. The transmission technologies broadcast and multicast differ because, in broadcast, the packet is transmitted to all hosts connected to the network. However, the packet is only transmitted to the intended recipients in a multicast.

In this article, you will learn about the difference between ***Broadcast*** and ***Multicast***. But before discussing the differences, you must know about Broadcast and Multicast.

What is broadcast?

Broadcast is a transmission mechanism that allows all network hosts to utilize the same communication channel. All other hosts receive a packet transmitted by a host on the network in broadcast mode. This method is also known as ***"all-to-all"***. A device on a ***Local Area Network (LAN)*** can be configured to broadcast messages to other devices on the network.

When a host broadcasts a packet, the intended recipient's address is specified in the packet's address field. Now that the packet has been broadcast, it is received by all other network sites. Each host examines the packet's address field after receiving it. If the packet contains the receiving host's address, it is handled by the receiving host. Otherwise, the packet is disregarded.

The broadcast function can address packets to all hosts on the network. To do this, the host transmitting the packet specifies a specific code in the packet's address field. When a packet with a specific code in the address field is sent, it is received and processed by every host in the network. A common example of broadcasting is a wireless network.

What is Multicast?

Multicasting is the simultaneous transmission of a message or set of packets to several networked devices. Multicasting is secured because only a select group of users receive the packets, and this technique reduces the necessary bandwidth because it distributes a single broadcast among several consumers. The most used protocol for multicasting is called ***User Datagram Protocol (UDP)***.

Several applications use multicasting. A user may send them to part of the recipients instead of emails to everyone on the contact list. Multicasting is also utilized in webcasts and video conferencing. Additionally, with the use of multicasting, devices like routers and switches deliver packets to a collection of connected devices.

Nowadays, multicasting has a variety of applications. For example, assessing a distributed database necessitates multicasting. The information in a distributed database is kept in multiple locations. As a result, the user's information request is broadcast to all database

locations, and the site has the intended information answers. Similarly, similar information is distributed to several customers in business, as is news distribution, teleconferencing, and distance learning.

Comparison between the Broadcast and Multicast

Here, you will learn the head-to-head comparisons between Broadcast and Multicast. The main differences between Broadcast and Multicast are as follows:

Features	Broadcast	Multicast
Definition	Broadcasting is a method of sending a message to all recipients simultaneously.	It is a group communication method in which data is sent simultaneously to a group of target computers.
Mapping	It contains one-to-all mapping.	It contains one-to-many mapping.
Bandwidth	The bandwidth of the broadcast is wasted.	The bandwidth of multicast is utilized effectively.
Management	It doesn't need any group management.	It needs group management to specify the group of hosts and stations which will receive packets.
Process	The bandwidth process is slow.	The multicast process is fast.
Traffic	It creates a large amount of network traffic by delivering each packet to every site on the network.	It keeps traffic under control by delivering packets only to interested hosts, lowering the network load.

LECTURE 4

What is Distance Vector Routing Algorithm?

Distance vector routing is a routing protocol that uses distance as a metric to determine the best path between two nodes. It is also known as the Bellman-Ford algorithm.

Distance vector routing is used in simple network topologies where the number of hops between two nodes is the primary metric used to determine the best path. In more complex network topologies, other factors such as link bandwidth and latency can be taken into account when determining the best path.

Advantages of Distance Vector Routing Algorithm

There major advantages to the distance vector routing algorithm are as follows:

- Distance vector routing is a routing protocol that uses the shortest path to a destination as its primary criterion. This algorithm is used in local-area networks, metropolitan-area networks, and wide-area networks.
- The main advantage of this algorithm is that it is easy to implement and does not require many resources.
- Additionally, distance vector routing converges quickly; it means that it can find an optimal route to a destination quickly after a change in network conditions.

Disadvantages of Distance Vector Routing Algorithm

There are a few disadvantages to the distance vector routing algorithm, as detailed below:

- First, it is not very scalable. This algorithm doesn't work well in large networks because the amount of information that needs to be exchanged between nodes can become too great.
- Additionally, this algorithm can suffer from routing loops. This occurs when there is a change in the network, and incorrect information is propagated between nodes, causing a loop.
- Also, this algorithm can be slow to converge on a new route after a change in the network.

Applications of Distance Vector Routing Algorithm

There are a number of different applications for the Distance Vector Routing Algorithm. One common application is in computer networking, specifically in routing data packets. This algorithm is also used in some types of telephone switching systems. Additionally, it has been used in military applications to route missiles.

Link State Routing;-

Link state routing is a method in which each router shares its neighbourhood's knowledge with every other router in the internetwork. In this algorithm, each router in the network understands the network topology then makes a routing table depend on this topology.

Each router will share data about its connection to its neighbour, who will, consecutively, reproduce the data to its neighbours, etc. This appears just before all routers have constructed a topology of the network.

In LSP, each node transmits its **IP address** and the **MAC** to its neighbour with its signature. Neighbours determine the signature and maintain a record of the combining IP address and the MAC. The Neighbour Lookup Protocol (NLP) of LSP derives and maintains the MAC and IP address of every network frame accepted by a node. The extracted data can support the mapping of MACs and IP addresses.

The link-state flooding algorithm prevents the general issues of broadcast in the existence of loops by having every node maintain a database of all LSP messages. The creator of each LSP contains its identity, data about the connection that has changed status, and also a sequence number.

The three keys to learning the Link State Routing algorithm are as follows −

Knowledge about the neighbourhood

Rather than transmitting its whole routing table, a router only shares data about its neighbourhood.

To all routers

Each router shares this data router on the network, not only its neighbours. This phase is known as flooding. Each router that receives the packet shares copies to all of its neighbours. Finally, every router (without exception) gets a copy of the comparable data. Information giving when there is a changeEach router assigns out data about the neighbours when there is a change. A route's value is used to every packet as it leaves a router and introduces a network in deciding a route.

Information giving when there is a change

This price is an outbound price, defining that it is used with a packet that leaves the router. Two factors govern how the price is used to packets in deciding a route. The price is used only by routers and not by some different stations on a network. The connection from one router to the next is a network, not a point-to-point cable. There are some topologies, including ring and bus, each station on the network determines each packet's header that passes.

Suppose the price was inserted by each station rather than the router alone. In that case, it could assemble unpredictably (the multiple stations in a network can transform for various circumstances, some of them unpredictable). A price is used as a packet that leaves the router instead of it being introduced. Some networks are transmitted. When a packet is in the network, each station containing the router can select it up. Therefore, we cannot create any price for the packet when it passes from a network to a router.

Congestion Control:-

What is **congestion**?
A state occurring in network layer when the message traffic is so heavy that it slows down network response time.

Effects of Congestion

- As delay increases, performance decreases.
- If delay increases, retransmission occurs, making situation worse.

Congestion control algorithms

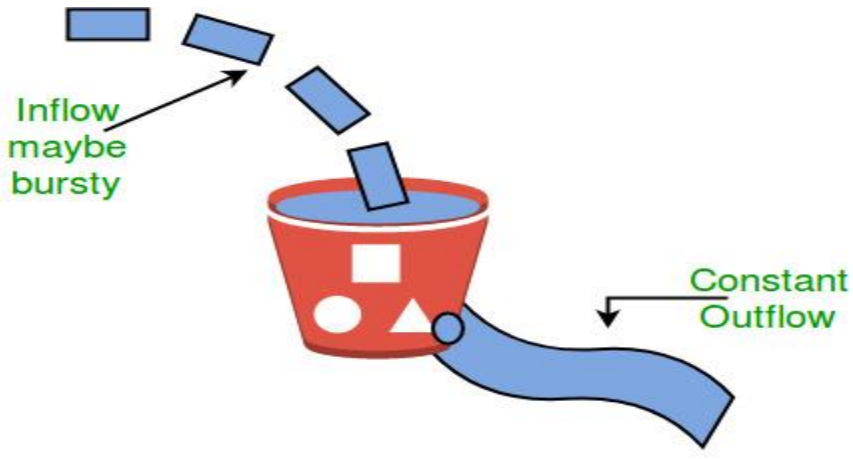
- Congestion Control is a mechanism that controls the entry of data packets into the network, enabling a better use of a shared network infrastructure and avoiding congestive collapse.
- Congestive-Avoidance Algorithms (CAA) are implemented at the TCP layer as the mechanism to avoid congestive collapse in a network.
- There are two congestion control algorithm which are as follows:

Leaky Bucket Algorithm

- The leaky bucket algorithm discovers its use in the context of network traffic shaping or rate-limiting.
- A leaky bucket execution and a token bucket execution are predominantly used for traffic shaping algorithms.
- This algorithm is used to control the rate at which traffic is sent to the network and shape the burst traffic to a steady traffic stream.
- The disadvantages compared with the leaky-bucket algorithm are the inefficient use of available network resources.
- The large area of network resources such as bandwidth is not being used effectively.

Let us consider an example to understand

Imagine a bucket with a small hole in the bottom.No matter at what rate water enters the bucket, the outflow is at constant rate.When the bucket is full with water additional water entering spills over the sides and is lost.



Similarly, each network interface contains a leaky bucket and the following **steps** are involved in leaky bucket algorithm:

1. When host wants to send packet, packet is thrown into the bucket.
2. The bucket leaks at a constant rate, meaning the network interface transmits packets at a constant rate.
3. Bursty traffic is converted to a uniform traffic by the leaky bucket.
4. In practice the bucket is a finite queue that outputs at a finite rate.

Token bucket Algorithm

- The leaky bucket algorithm has a rigid output design at an average rate independent of the bursty traffic.
- In some applications, when large bursts arrive, the output is allowed to speed up. This calls for a more flexible algorithm, preferably one that never loses information. Therefore, a token bucket algorithm finds its uses in network traffic shaping or rate-limiting.
- It is a control algorithm that indicates when traffic should be sent. This order comes based on the display of tokens in the bucket.
- The bucket contains tokens. Each of the tokens defines a packet of predetermined size. Tokens in the bucket are deleted for the ability to share a packet.
- When tokens are shown, a flow to transmit traffic appears in the display of tokens.
- No token means no flow sends its packets. Hence, a flow transfers traffic up to its peak burst rate in good tokens in the bucket.

Need of token bucket Algorithm:-

The leaky bucket algorithm enforces output pattern at the average rate, no matter how bursty the traffic is. So in order to deal with the bursty traffic we need a flexible algorithm so that the data is not lost. One such algorithm is token bucket algorithm.

Steps of this algorithm can be described as follows:

1. In regular intervals tokens are thrown into the bucket.
2. The bucket has a maximum capacity.
3. If there is a ready packet, a token is removed from the bucket, and the packet is sent.
4. If there is no token in the bucket, the packet cannot be sent.

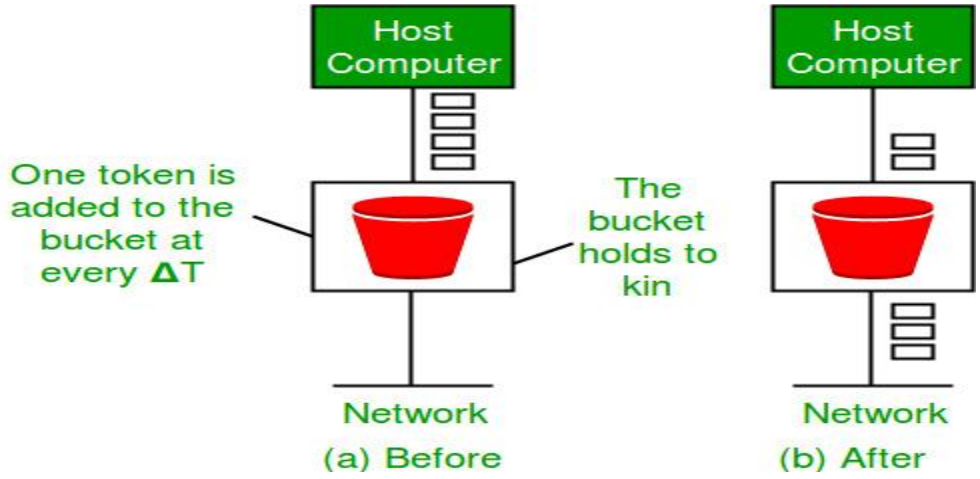
Let’s understand with an example,

In figure (A) we see a bucket holding three tokens, with five packets waiting to be transmitted. For a packet to be transmitted, it must capture and destroy one token. In figure (B) We see that three of the five packets have gotten through, but the other two are stuck waiting for more tokens to be generated.

Ways in which token bucket is superior to leaky bucket: The leaky bucket algorithm controls the rate at which the packets are introduced in the network, but it is very conservative in nature. Some flexibility is introduced in the token bucket algorithm. In the token bucket, algorithm tokens are generated at each tick (up to a certain limit). For an incoming packet to be transmitted, it must capture a token and the transmission takes place at the same rate. Hence some of the busty packets are transmitted at the same rate if tokens are available and thus introduces some amount of flexibility in the system.

Formula: M * s = C + ? * s where S – is time taken M – Maximum output rate ? – Token arrival rate C – Capacity of the token bucket in byte

Let’s understand with an example,



LECTURE 5

What is QoS in Networking?

Quality of service (QoS) is the use of mechanisms or technologies that work on a network to control traffic and ensure the performance of critical applications with limited network capacity. It enables organizations to adjust their overall network traffic by prioritizing specific high-performance applications.

QoS is typically applied to networks that carry traffic for resource-intensive systems. Common services for which it is required include internet protocol television (IPTV), online gaming, streaming media, videoconferencing, video on demand (VOD), and Voice over IP (VoIP).

Using QoS in networking, organizations have the ability to optimize the performance of multiple applications on their network and gain visibility into the bit rate, delay, jitter, and packet rate of their network. This ensures they can engineer the traffic on their network and change the way that packets are routed to the internet or other networks to avoid transmission delay. This also ensures that the organization achieves the expected service quality for applications and delivers expected user experiences.

As per the QoS meaning, the key goal is to enable networks and organizations to prioritize traffic, which includes offering dedicated bandwidth, controlled jitter, and lower latency. The technologies used to ensure this are vital to enhancing the performance of business applications, wide-area networks (WANs), and service provider networks.

How Does QoS Work?

QoS networking technology works by marking packets to identify service types, then configuring routers to create separate virtual queues for each application, based on their priority. As a result, bandwidth is reserved for critical applications or websites that have been assigned priority access.

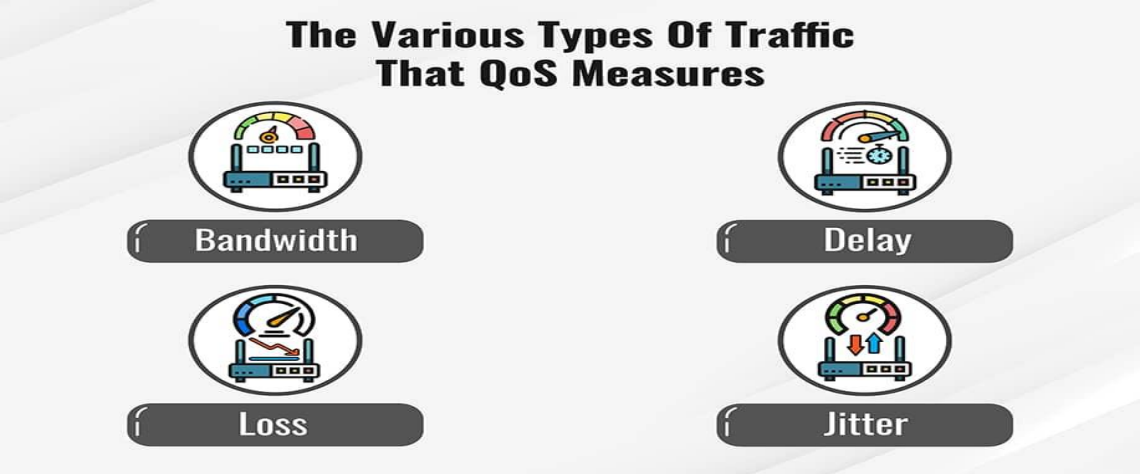
QoS technologies provide capacity and handling allocation to specific flows in network traffic. This enables the network administrator to assign the order in which packets are handled and provide the appropriate amount of bandwidth to each application or traffic flow.

Types of Network Traffic

Understanding how QoS network software works is reliant on defining the various types of traffic that it measures. These are:

1. Bandwidth: The speed of a link. QoS can tell a router how to use bandwidth. For example, assigning a certain amount of bandwidth to different queues for different traffic types.
2. Delay: The time it takes for a packet to go from its source to its end destination. This can often be affected by queuing delay, which occurs during times of congestion and a packet waits in a queue before being transmitted. QoS enables organizations to avoid this by creating a priority queue for certain types of traffic.
3. Loss: The amount of data lost as a result of packet loss, which typically occurs due to network congestion. QoS enables organizations to decide which packets to drop in this event.

4. Jitter: The irregular speed of packets on a network as a result of congestion, which can result in packets arriving late and out of sequence. This can cause distortion or gaps in audio and video being delivered.



Implementing QoS begins with an enterprise identifying the types of traffic that are important to them, use high volumes of bandwidth, and/or are sensitive to latency or packet loss.

This helps the organization understand the needs and importance of each traffic type on its network and design an overall approach. For example, some organizations may only need to configure bandwidth limits for specific services, whereas others may need to fully configure interface and security policy bandwidth limits for all their services, as well as prioritize queuing critical services relative to traffic rate.

The organization can then deploy policies that classify traffic and ensure the availability and consistency of its most important applications. Traffic can be classified by port or internet protocol (IP), or through a more sophisticated approach such as by application or user.

Bandwidth management and queuing tools are then assigned roles to handle traffic flow specifically based on the classification they received when they entered the network. This allows for packets within traffic flows to be stored until the network is ready to process them. Priority queuing can also be used to ensure the necessary availability and minimal latency of network performance for important applications and traffic. This is so that the network's most important activities are not starved of bandwidth by those of lesser priority.

Furthermore, bandwidth management measures and controls traffic flow on the network infrastructure to ensure it does not exceed capacity and prevent congestion. This includes using traffic shaping, a rate-limiting technique that optimizes or guarantees performance and increases usable bandwidth, and scheduling algorithms, which offer several methods for providing bandwidth to specific traffic flows.

Why is QoS Important?

Traditional business networks operated as separate entities. Phone calls and teleconferences were handled by one network, while laptops, desktops, servers and other devices connected to

another. They rarely crossed paths, unless a computer used a telephone line to access the internet.

When networks only carried data, speed was not overly critical. But now, interactive applications carrying audio and video content need to be delivered at high speed, without packet loss or variations in delivery speed.

QoS is particularly important to guarantee the high performance of critical applications that require high bandwidth for real-time traffic. For example, it helps businesses to prioritize the performance of "inelastic" applications that often have minimum bandwidth requirements, maximum latency limits, and high sensitivity to jitter and latency, such as VoIP and videoconferencing.

QoS helps businesses prevent the delay of these sensitive applications, ensuring they perform to the level that users require. For example, lost packets could cause a delay to the stream, which results in the sound and video quality of a videoconference call to become choppy and indecipherable.

QoS is increasingly important as network performance requirements adapt to the growing number of people using them. The latest online applications and services require vast amounts of bandwidth and network performance, and users demand they offer high performance at all times. Organizations, therefore, need to deploy techniques and technologies that guarantee the best possible service.

QoS is also becoming increasingly important as the Internet of Things (IoT) continues to come to maturity. For example, in the manufacturing sector, machines now leverage networks to provide real-time status updates on any potential issues. Therefore, any delay in feedback could cause highly costly mistakes in IoT networking. QoS enables the data stream to take priority in the network and ensures that the information flows as quickly as possible.

Cities are now filled with smart sensors that are vital to running large-scale IoT projects such as smart buildings. The data collected and analyzed, such as humidity and temperature data, is often highly time-sensitive and needs to be identified, marked, and queued appropriately.

Advantages of QoS

The deployment of QoS is crucial for businesses that want to ensure the availability of their business-critical applications. It is vital for delivering differentiated bandwidth and ensuring data transmission takes place without interrupting traffic flow or causing packet losses. Major advantages of deploying QoS include:

1. **Unlimited application prioritization:** QoS guarantees that businesses' most mission-critical applications will always have priority and the necessary resources to achieve high performance.
2. **Better resource management:** QoS enables administrators to better manage the organization's internet resources. This also reduces costs and the need for investments in link expansions.
3. **Enhanced user experience:** The end goal of QoS is to guarantee the high performance of critical applications, which boils down to delivering optimal user experience.

Employees enjoy high performance on their high-bandwidth applications, which enables them to be more effective and get their job done more quickly.

4. **Point-to-point traffic management:** Managing a network is vital however traffic is delivered, be it end to end, node to node, or point to point. The latter enables organizations to deliver customer packets in order from one point to the next over the internet without suffering any packet loss.
5. **Packet loss prevention:** Packet loss can occur when packets of data are dropped in transit between networks. This can often be caused by a failure or inefficiency, network congestion, a faulty router, loose connection, or poor signal. QoS avoids the potential of packet loss by prioritizing bandwidth of high-performance applications.
6. **Latency reduction:** Latency is the time it takes for a network request to go from the sender to the receiver and for the receiver to process it. This is typically affected by routers taking longer to analyze information and storage delays caused by intermediate switches and bridges. QoS enables organizations to reduce latency, or speed up the process of a network request, by prioritizing their critical application.

INTERNETWORKING

In internetworking, routers are aware of each other's addresses as well as addresses outside of their network. They have two options: they can be statically set to operate on various networks, or they can learn by utilising the internetworking routing protocol.

What is Internetworking in a Computer Network?

The word "internetworking," which combines the words "inter" and "networking," denotes a connection between completely distinct nodes/segments. This connection is made possible by intermediary hardware like routers or gateways. Catenet was the initial title for associate degree internetwork. Private, public, commercial, industrial, and governmental networks frequently connect to one another. Therefore, a degree of internetwork could be a collection of several networks that operate as a single large network and are connected by intermediate networking devices. The trade, goods, and methods used to address the difficulty of creating and managing internet works are referred to as internetworking.

How Does Internetworking Work?

Every network node or phase is built using a similar protocol or a communication logic, such as TCP (Transfer Control Protocol) or IP (Internet Protocol), to enable communication. It is referred to as "internetworking" when a network interacts with another network using ongoing communication protocols. A packet of information must be delivered across numerous links, which is a problem that internetworking was created to address.

The distinction between expanding the network and internetworking is quite slight. A simple extension of a LAN is the use of a switch or hub to join two local area networks, but connecting them via a router is an example of internetworking. The OSI-ISO model's Layer 3 (Network Layer) enforces internetworking. The internet is the most prominent famous example of internetworking.

Types of Internetworking

Internetworking primarily consists of three units: Extranet, Internet, and Intranet. Internet connections may or may not be present on intranets and extranets. The computer network or the extranet area unit is typically protected from being accessed from the internet if it is not approved and if there is a link to the internet. Although it should serve as a portal for access to portions of the associate degree extranet, the internet is not considered to be a part of the computer network or extranet.

Extranet

It's a network of the internetwork with a confined scope to one organisation or institution but with limited links to one or more other networks on occasion; however, this is not always the case. It is the lowest degree of internet usage and is typically prohibited in extremely private areas. An extranet may also be referred to as a MAN, WAN, or another type of network, but it cannot include a single local area network; rather, it must make at least one mention of an external network.

Internet

Internet is a specific internetworking that connects governmental, academic, public, and private networks on a global scale. It is based on the ARPANET, which was created by the ARPA (Advanced Research Projects Agency) of the U.S. Defense Department. It is also the location of the World Wide Web (WWW) and is referred to as the "Internet" to distinguish it from other generic internetworking. Internet users and their service providers utilise IP addresses obtained from address registries that control assignments.

Intranet

This computer network can be a collection of interconnected networks that employ the Internet Protocol and IP-based software like web browsers as well as FTP tools, all of which are controlled by a single body entity. This body entity blocks access to the computer network for the rest of the world and only allows a select few users. This network most frequently refers to the internal network of a business or other enterprise. To provide users with browseable data, a large computer network can typically have its own internet server.

Why Internetworking?

A few major issues, including duplicated resources, separated LANs, and a lack of network administration, have led to the evolution of internetworking. Transmission issues across completely distinct offices or departments were caused by isolated LANs. Duplication of resources required distinct support staff and continual hardware and code to be delivered to each business or department. Due to the absence of network administration, there was no centralised system for managing or troubleshooting networks.

At the link layer of a networking model, which is the hardware-centric layer present below the number of TCP/IP logical interfaces, another type of network interconnection typically occurs between businesses. Network switches and bridges are used to connect the various networks. However, the resulting system is essentially a larger, single subnetwork, no

internetworking protocol, such as web protocol, is required to traverse these devices. It is often wrongly referred to as "internetworking."

However, by segmenting the network into phases and then logically splitting the segment traffic using routers, one electronic network can also be reborn into an associated internetwork. The Internet Protocol is designed to provide a network with an unstable packet service. The approach avoids using intermediate network components to keep the network in any state. Instead, the endpoints of each communication session are given this task. Applications should use some suitable Transport Layer protocol, like TCP, which offers a dependable stream, in order to send information correctly. User Datagram Protocol (UDP), a less complex, connection-free transport protocol, is used by some apps for tasks like voice chat and video streaming that don't require timely or accurate information transmission.

Fragmentation:-

- **Fragmentation** is done by the network layer when the maximum size of datagram is greater than maximum size of data that can be held in a frame i.e., its Maximum Transmission Unit (MTU). The network layer divides the datagram received from the transport layer into fragments so that data flow is not disrupted.

- Since there are 16 bits for total length in IP header so, the maximum size of IP datagram = 2^{16} – 1 = 65, 535 bytes.

- It is done by the network layer at the destination side and is usually done at routers.
- Source side does not require fragmentation due to wise (good) segmentation by transport layer i.e. instead of doing segmentation at the transport layer and fragmentation at the network layer, the transport layer looks at datagram data limit and frame data limit and does segmentation in such a way that resulting data can easily fit in a frame without the need of fragmentation.
- Receiver identifies the frame with the **identification (16 bits)** field in the IP header. Each fragment of a frame has the same identification number.
- Receiver identifies the sequence of frames using the **fragment offset(13 bits)** field in the IP header
- Overhead at the network layer is present due to the extra header introduced due to fragmentation.

The need of Fragmentation at Network Layer:

Fragmentation at the Network Layer is a process of dividing a large data packet into smaller pieces, known as fragments, to improve the efficiency of data transmission over a network. The need for fragmentation at the network layer arises from several factors:

1. Maximum Transmission Unit (MTU): Different networks have different Maximum Transmission Unit (MTU) sizes, which determine the maximum size of a data packet that can be transmitted over that network. If the size of a data packet exceeds the MTU, it needs to be fragmented into smaller fragments that can be transmitted over the network.
2. Network Performance: Large data packets can consume a significant amount of network resources and can cause congestion in the network. Fragmentation helps to reduce the impact

of large data packets on network performance by breaking them down into smaller fragments that can be transmitted more efficiently.
3. Bandwidth Utilization: Large data packets may consume a significant amount of network bandwidth, causing other network traffic to be slowed down. Fragmentation helps to reduce the impact of large data packets on network bandwidth utilization by breaking them down into smaller fragments that can be transmitted more efficiently.

Fragmentation at the network layer is necessary in order to ensure efficient and reliable transmission of data over communication networks.

1. Large Packet Size: In some cases, the size of the packet to be transmitted may be too large for the underlying communication network to handle. Fragmentation at the network layer allows the large packet to be divided into smaller fragments that can be transmitted over the network.
2. Path MTU: The Maximum Transmission Unit (MTU) of a network defines the largest packet size that can be transmitted over the network. Fragmentation at the network layer allows the packet to be divided into smaller fragments that can be transmitted over networks with different MTU values.
3. Reliable Transmission: Fragmentation at the network layer increases the reliability of data transmission, as smaller fragments are less likely to be lost or corrupted during transmission.

Fields in IP header for fragmentation –

- **Identification (16 bits) –** use to identify fragments of the same frame.
- **Fragment offset (13 bits) –** use to identify the sequence of fragments in the frame. It generally indicates a number of data bytes preceding or ahead of the fragment. Maximum fragment offset possible = (65535 – 20) = 65515 {where 65535 is the maximum size of datagram and 20 is the minimum size of IP header}
  So, we need ceil($\log_2 65515$) = 16 bits for a fragment offset but the fragment offset field has only 13 bits. So, to represent efficiently we need to scale down the fragment offset field by $2^{16}/2^{13} = 8$ which acts as a scaling factor. Hence, all fragments except the last fragment should have data in multiples of 8 so that fragment offset $\in$ N.
- **More fragments (MF = 1 bit) –** tells if more fragments are ahead of this fragment i.e. if MF = 1, more fragments are ahead of this fragment and if MF = 0, it is the last fragment.
- **Don't fragment (DF = 1 bit) –** if we don't want the packet to be fragmented then DF is set i.e. DF = 1.

Reassembly of Fragments –

It takes place only at the destination and not at routers since packets take an independent path(datagram packet switching), so all may not meet at a router and hence a need of fragmentation may arise again. The fragments may arrive out of order also.

MF	Fragment Offset	
1	0	→ 1st packet
1	!=0	→ Intermediate packet
0	!=0	→ Last packet
0	0	→ Invalid

LECTURE 6

Internet Layer

The Internet layer, also known as the **network layer** or **IP layer**, accepts and delivers packets for the network. This layer includes the powerful Internet Protocol (IP), the Address Resolution Protocol (ARP), and the Internet Control Message Protocol (ICMP).

IP Protocol

The IP protocol and its associated routing protocols are possibly the most significant of the entire TCP/IP suite. IP is responsible for the following:

- **IP addressing** – The IP addressing conventions are part of the IP protocol. Designing an IPv4 Addressing Scheme introduces IPv4 addressing and IPv6 Addressing Overview introduces IPv6 addressing.
- **Host-to-host communications** – IP determines the path a packet must take, based on the receiving system's IP address.
- **Packet formatting** – IP assembles packets into units that are known as **datagram's**. Datagram's are fully described in Internet Layer: Where Packets Are Prepared for Delivery.
- **Fragmentation** – If a packet is too large for transmission over the network media, IP on the sending system breaks the packet into smaller fragments. IP on the receiving system then reconstructs the fragments into the original packet.

Oracle Solaris supports both IPv4 and IPv6 addressing formats, which are described in this book. To avoid confusion when addressing the Internet Protocol, one of the following conventions is used:

- When the term "IP" is used in a description, the description applies to both IPv4 and IPv6.
- When the term "IPv4" is used in a description, the description applies only to IPv4.
- When the term "IPv6" is used in a description, the description applies only to IPv6.

ARP Protocol

The Address Resolution Protocol (ARP) conceptually exists between the data-link and Internet layers. ARP assists IP in directing datagrams to the appropriate receiving system by mapping Ethernet addresses (48 bits long) to known IP addresses (32 bits long).

ICMP Protocol

The Internet Control Message Protocol (ICMP) detects and reports network error conditions. ICMP reports on the following:

- **Dropped packets** – Packets that arrive too fast to be processed
- **Connectivity failure** – A destination system cannot be reached
- **Redirection** – Redirecting a sending system to use another router

IP address definition

An IP address is a unique address that identifies a device on the internet or a local network. IP stands for "Internet Protocol," which is the set of rules governing the format of data sent via the internet or local network.

In essence, IP addresses are the identifier that allows information to be sent between devices on a network: they contain location information and make devices accessible for communication. The internet needs a way to differentiate between different computers, routers, and websites. IP addresses provide a way of doing so and form an essential part of how the internet works.

What is an IP Address?

An IP address is a string of numbers separated by periods. IP addresses are expressed as a set of four numbers — an example address might be 192.158.1.38. Each number in the set can range from 0 to 255. So, the full IP addressing range goes from 0.0.0.0 to 255.255.255.255.

IP addresses are not random. They are mathematically produced and allocated by the Internet Assigned Numbers Authority (IANA), a division of the Internet Corporation for Assigned Names and Numbers (ICANN). ICANN is a non-profit organization that was established in the United States in 1998 to help maintain the security of the internet and allow it to be usable by all. Each time anyone registers a domain on the internet, they go through a domain name registrar, who pays a small fee to ICANN to register the domain.

Watch this video to learn what IP address is, why IP address is important and how to protect it from hackers:

How do IP addresses work

If you want to understand why a particular device is not connecting in the way you would expect or you want to troubleshoot why your network may not be working, it helps understand how IP addresses work.

Internet Protocol works the same way as any other language, by communicating using set guidelines to pass information. All devices find, send, and exchange information with other connected devices using this protocol. By speaking the same language, any computer in any location can talk to one another.

The use of IP addresses typically happens behind the scenes. The process works like this:

1. **Your device indirectly connects to the internet by connecting at first to a network connected to the internet, which then grants your device access to the internet.**
2. **When you are ~~at~~ home, that network will probably be your Internet Service Provider (ISP). At work, it will be your company network.**
3. **Your IP address is assigned to your device by your ISP.**
4. Your internet activity goes through the ISP, and they route it back to you, using your IP address. Since they are giving you access to the internet, it is their role to assign an IP address to your device.
5. However, your IP address can change. For example, turning your modem or router on or off can change it. Or you can contact your ISP, and they can change it for you.
6. When you are out and about – for example, traveling – and you take your device with you, your home IP address does not come with you. This is because you will be using another network (Wi-Fi at a hotel, airport, or coffee shop, etc.) to access the internet and will be using a different (and temporary) IP address, assigned to you by the ISP of the hotel, airport or coffee shop.

As the process implies, there are different types of IP addresses, which we explore below.

Types of IP addresses

There are different categories of IP addresses, and within each category, different types.

Consumer IP addresses

Every individual or business with an internet service plan ~~will~~ have two types of IP addresses: their private IP addresses and their public IP address. The terms public and private relate to the network location — that is, a private IP address is used inside a network, while a public one is used outside a network.

Private IP addresses

Every device that connects to your internet network has a private IP address. This includes computers, smart phones, and tablets but also any Bluetooth-enabled devices like speakers, printers, or smart TVs. With the growing internet of things, the number of private IP addresses you have at home is probably growing. Your router needs a way to identify these items separately, and many items need a way to recognize each other.

Therefore, your router generates private IP addresses that are unique identifiers for each device that differentiate them on the network.

Public IP addresses

A public IP address is the primary address associated with your whole network. While each connected device has its own IP address, they are also included within the main IP address for your network. As described above, your public IP address is provided to your router by your ISP. Typically, ISPs have a large pool of IP addresses that they distribute to their customers. Your public IP address is the address that all the devices outside your internet network will use to recognize your network.

Public IP addresses

Public IP addresses come in two forms – dynamic and static.

Dynamic IP addresses

Dynamic IP addresses change automatically and regularly. ISPs buy a large pool of IP addresses and assign them automatically to their customers. Periodically, they re-assign them and put the older IP addresses back into the pool to be used for other customers. The rationale for this approach is to generate cost savings for the ISP. Automating the regular movement of IP addresses means they don't have to carry out specific actions to re-establish a customer's IP address if they move home, for example. There are security benefits, too, because a changing IP address makes it harder for criminals to hack into your network interface.

Static IP addresses

In contrast to dynamic IP addresses, static addresses remain consistent. Once the network assigns an IP address, it remains the same. Most individuals and businesses do not need a static IP address, but for businesses that plan to host their own server, it is crucial to have one. This is because a static IP address ensures that websites and email addresses tied to it will have a consistent IP address — vital if you want other devices to be able to find them consistently on the web.

This leads to the next point – which is the two types of website IP addresses.

There are two types of website IP addresses

For website owners who don't host their own server, and instead rely on a web hosting package – which is the case for most websites – there are two types of website IP addresses. These are shared and dedicated.

Shared IP addresses

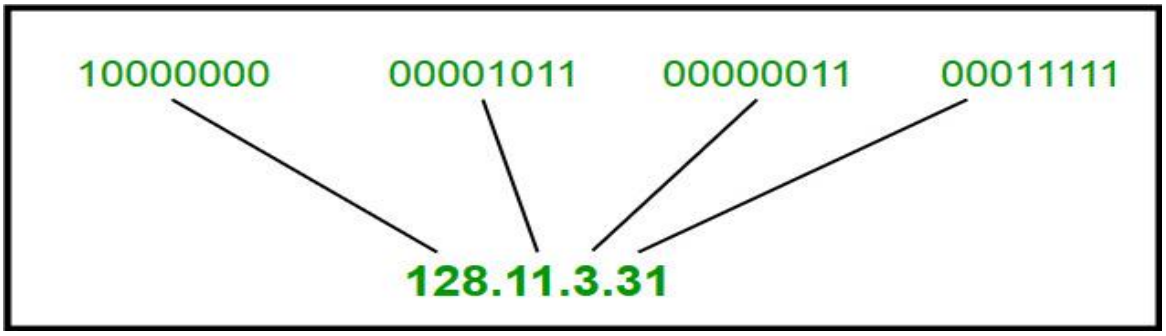
Websites that rely on shared hosting plans from web hosting providers will typically be one of many websites hosted on the same server. This tends to be the case for individual websites or SME websites, where traffic volumes are manageable, and the sites themselves are limited in terms of the number of pages, etc. Websites hosted in this way will have shared IP addresses.

Dedicated IP addresses

Some web hosting plans have the option to purchase a dedicated IP address (or addresses). This can make obtaining an SSL certificate easier and allows you to run your own File Transfer Protocol (FTP) server. This makes it easier to share and transfer files with multiple people within an organization and allow anonymous FTP sharing options. A dedicated IP address also allows you to access your website using the IP address alone rather than the domain name — useful if you want to build and test it before registering your domain.

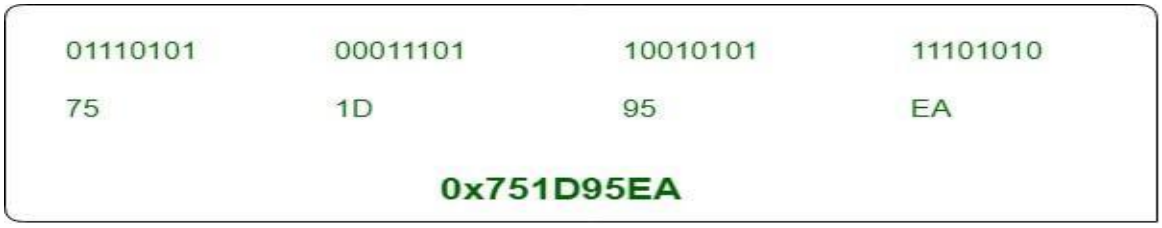
An IP address is an address having information about how to reach a specific host, especially outside the LAN. An IP address is a 32-bit unique address having an address space of 232.
Generally, there are two notations in which the IP address is written, dotted decimal notation and hexadecimal notation.

Dotted Decimal Notation



Dotted Decimal Notation

Hexadecimal Notation



Some points to be noted about dotted decimal notation:

1. The value of any segment (byte) is between 0 and 255 (both included).
2. No zeroes are preceding the value in any segment (054 is wrong, 54 is correct).

Classful Addressing

The 32-bit IP address is divided into five sub-classes. These are:

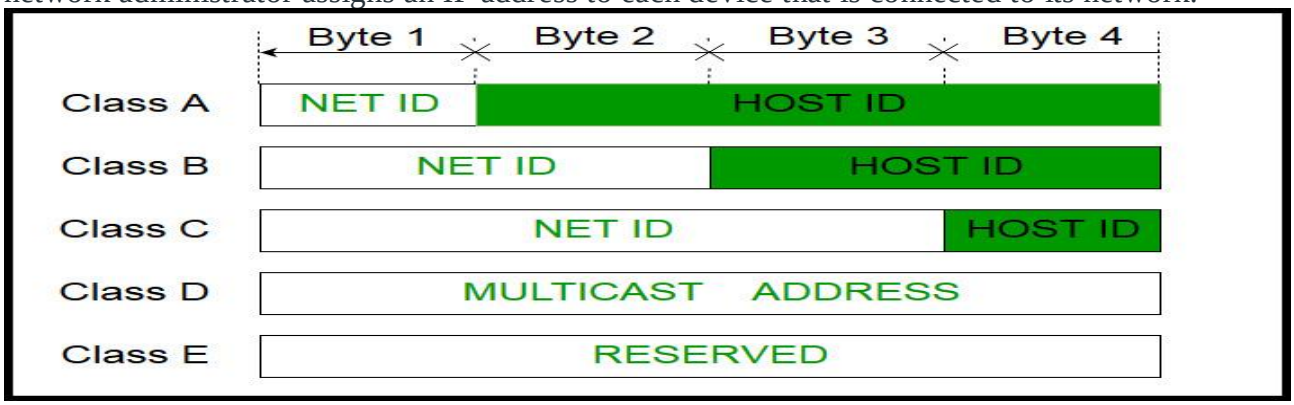
- Class A
- Class B
- Class C
- Class D
- Class E

Each of these classes has a valid range of IP addresses. Classes D and E are reserved for multicast and experimental purposes respectively. The order of bits in the first octet

determines the classes of the IP address.
The IPv4 address is divided into two parts:

- Network ID
- Host ID

The class of IP address is used to determine the bits used for network ID and host ID and the number of total networks and hosts possible in that particular class. Each ISP or network administrator assigns an IP address to each device that is connected to its network.



Classful Addressing

Note:

1. IP addresses are globally managed by Internet Assigned Numbers Authority(IANA) and regional Internet registries(RIR).
2. While finding the total number of host IP addresses, 2 IP addresses are not counted and are therefore, decreased from the total count because the first IP address of any network is the network number and whereas the last IP address is reserved for broadcast IP.

Class A

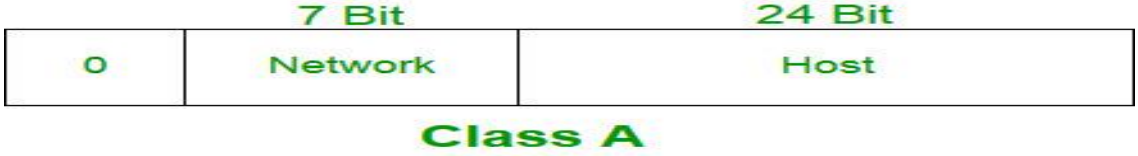
IP addresses belonging to class A are assigned to the networks that contain a large number of hosts.

- The network ID is 8 bits long.
- The host ID is 24 bits long.

The higher-order bit of the first octet in class A is always set to 0. The remaining 7 bits in the first octet are used to determine network ID. The 24 bits of host ID are used to determine the host in any network. The default subnet mask for Class A is 255.x.x.x. Therefore, class A has a total of:

- 2^24 – 2 = 16,777,214 host ID

IP addresses belonging to class A ranges from 1.0.0.0 – 126.255.255.255.



Class B

IP address belonging to class B is assigned to networks that range from medium-sized to large-sized networks.

- The network ID is 16 bits long.

- The host ID is 16 bits long.

The higher-order bits of the first octet of IP addresses of class B are always set to 10. The remaining 14 bits are used to determine the network ID. The 16 bits of host ID are used to determine the host in any network. The default subnet mask for class B is 255.255.x.x. Class B has a total of:

- 2^14 = 16384 network address
- 2^16 – 2 = 65534 host address



IP addresses belonging to class B ranges from 128.0.0.0 – 191.255.255.255.

Class B

Class C

IP addresses belonging to class C are assigned to small-sized networks.

- The network ID is 24 bits long.
- The host ID is 8 bits long.

The higher-order bits of the first octet of IP addresses of class C is always set to 110. The remaining 21 bits are used to determine the network ID. The 8 bits of host ID are used to determine the host in any network. The default subnet mask for class C is 255.255.255.x. Class C has a total of:

- 2^21 = 2097152 network address
- 2^8 – 2 = 254 host address

IP addresses belonging to class C range from 192.0.0.0 – 223.255.255.255.



Class C

Class D

IP address belonging to class D is reserved for multi-casting. The higher-order bits of the first octet of IP addresses belonging to class D is always set to 1110. The remaining bits are for the address that interested hosts recognize.

Class D does not possess any subnet mask. IP addresses belonging to class D range from 224.0.0.0 – 239.255.255.255.



Class D

Class E

IP addresses belonging to class E are reserved for experimental and research purposes. IP addresses of class E range from 240.0.0.0 – 255.255.255.254. This class doesn't have any subnet mask. The higher-order bits of the first octet of class E are always set to 1111.



Class E

Range of Special IP Addresses

169.254.0.0 – 169.254.0.16 : Link-local addresses
127.0.0.0 – 127.255.255.255 : Loop-back addresses
0.0.0.0 – 0.0.0.8: used to communicate within the current network.

Summary of Classful Addressing

CLASS	LEADING BITS	NET ID BITS	HOST ID BITS	NO. OF NETWORKS	ADDRESSES PER NETWORK	START ADDRESS	END ADDRESS
CLASS A	0	8	24	2^7 (128)	2^{24} (16,777,216)	0.0.0.0	127.255.255.255
CLASS B	10	16	16	2^{14} (16,384)	2^{16} (65,536)	128.0.0.0	191.255.255.255
CLASS C	110	24	8	2^{21} (2,097,152)	2^8 (256)	192.0.0.0	223.255.255.255
CLASS D	1110	NOT DEFINED	NOT DEFINED	NOT DEFINED	NOT DEFINED	224.0.0.0	239.255.255.255
CLASS E	1111	NOT DEFINED	NOT DEFINED	NOT DEFINED	NOT DEFINED	240.0.0.0	255.255.255.255

LECTURE 7

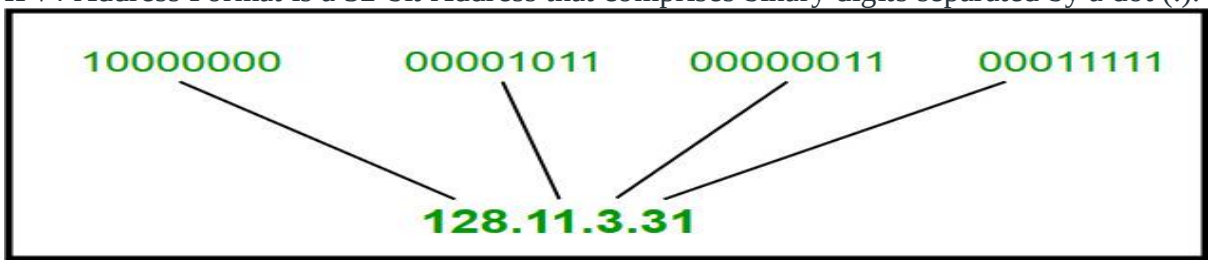
IPv4

IPv4 address consists of two things that are the network address and the host address. It stands for **Internet Protocol version four.** It was introduced in 1981 by DARPA and was the first deployed version in 1982 for production on SATNET and on the ARPANET in January 1983.

IPv4 addresses are 32-bit integers that have to be expressed in Decimal Notation. It is represented by 4 numbers separated by dots in the range of 0-255, which have to be converted to 0 and 1, to be understood by Computers. For Example, An IPv4 Address can be written as **189.123.123.90.**

IPv4 Address Format

IPv4 Address Format is a 32-bit Address that comprises binary digits separated by a dot (.).



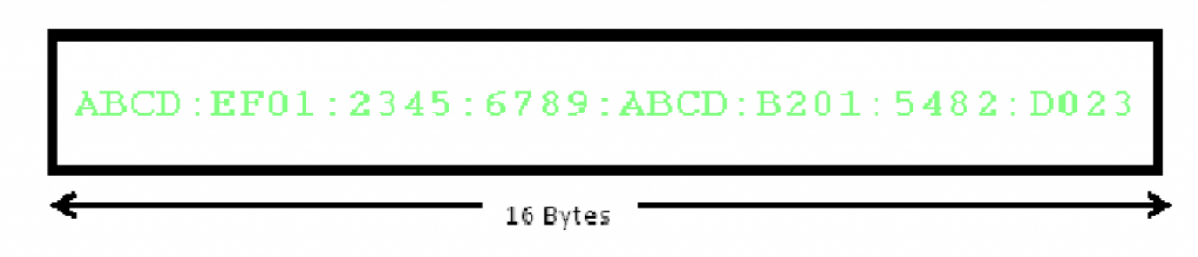
IPv4 Address Format

IPv6

IPv6 is based on IPv4 and stands for Internet Protocol version 6. It was first introduced in December 1995 by Internet Engineering Task Force. IP version 6 is the new version of Internet Protocol, which is way better than IP version 4 in terms of complexity and efficiency. IPv6 is written as a group of 8 hexadecimal numbers separated by colon (:). It can be written as 128 bits of 0s and 1s.

IPv6 Address Format

IPv6 Address Format is a 128-bit IP Address, which is written in a group of 8 hexadecimal numbers separated by colon (:).



IPv6 Address Format

Benefits of IPv6

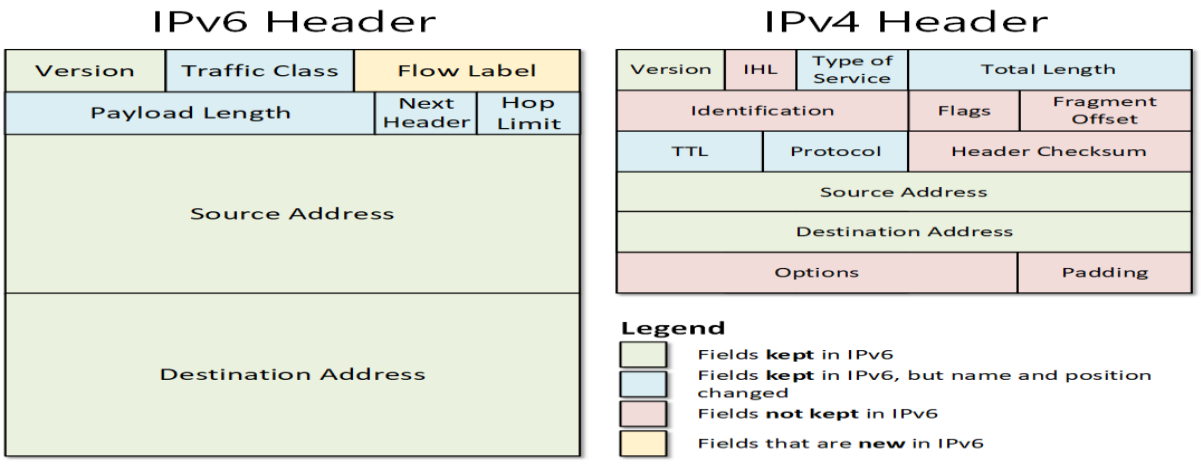
The recent Version of IP IPv6 has a greater advantage over IPv4. Here are some of the mentioned benefits:

- **Larger Address Space:** IPv6 has a greater address space than IPv4, which is required for expanding the IP Connected Devices. IPv6 has 128 bit IP Address rather and IPv4 has a 32-bit Address.
- **Improved Security:** IPv6 has some improved security which is built in with it. IPv6 offers security like Data Authentication, Data Encryption, etc. Here, an Internet Connection is more Secure.
- **Simplified Header Format:** As compared to IPv4, IPv6 has a simpler and more effective header Structure, which is more cost-effective and also increases the speed of Internet Connection.
- **Prioritize:** IPv6 contains stronger and more reliable support for QoS features, which helps in increasing traffic over websites and increases audio and video quality on pages.
- **Improved Support for Mobile Devices:** IPv6 has increased and better support for Mobile Devices. It helps in making quick connections over other Mobile Devices and in a safer way than IPv4.

Difference Between IPv4 and IPv6

IPv4	IPv6
IPv4 has a 32-bit address length	IPv6 has a 128-bit address length
It Supports Manual and DHCP address configuration	It supports Auto and renumbering address configuration
In IPv4 end to end, connection integrity is Unachievable	In IPv6 end-to-end, connection integrity is Achievable
It can generate 4.29×109 address space	The address space of IPv6 is quite large it can produce 3.4×1038 address space
The Security feature is dependent on the application	IPSEC is an inbuilt security feature in the IPv6 protocol
Address representation of IPv4 is in decimal	Address Representation of IPv6 is in hexadecimal

IPv4	IPv6
Fragmentation performed by Sender and forwarding routers	In IPv6 fragmentation is performed only by the sender
In IPv4 Packet flow identification is not available	In IPv6 packet flow identification are Available and uses the flow label field in the header
In IPv4 checksum field is available	In IPv6 checksum field is not available
It has a broadcast Message Transmission Scheme	In IPv6 multicast and anycast message transmission scheme is available
In IPv4 Encryption and Authentication facility not provided	In IPv6 Encryption and Authentication are provided
IPv4 has a header of 20-60 bytes.	IPv6 has a header of 40 bytes fixed
IPv4 can be converted to IPv6	Not all IPv6 can be converted to IPv4
IPv4 consists of 4 fields which are separated by addresses dot (.)	IPv6 consists of 8 fields, which are separated by a colon (:)
IPv4's IP addresses are divided into five different classes. Class A , Class B, Class C, Class D , Class E.	IPv6 does not have any classes of the IP address.
IPv4 supports VLSM(Variable Length subnet mask).	IPv6 does not support VLSM.
Example of IPv4: 66.94.29.13	Example of IPv6: 2001:0000:3238:DFE1:0063:0000:0000:FEFB



What is CIDR?

Classless Inter-Domain Routing (CIDR) is an IP address allocation method that improves data routing efficiency on the internet. Every machine, server, and end-user device that connects to the internet has a unique number, called an IP address, associated with it. Devices find and communicate with one another by using these IP addresses. Organizations use CIDR to allocate IP addresses flexibly and efficiently in their networks.

What are the different IP address formats?

An IP address has two parts:

- The *network address* is a series of numerical digits pointing to the network's unique identifier
- The *host address* is a series of numbers indicating the host or individual device identifier on the network

Until the early 1990s, IP addresses were allocated using the classful addressing system. The total length of the address was fixed, and the number of bits allocated to the network and host portions were also fixed.

Classful addresses

An IPv4 address consists of 32 bits. Each string of numbers separated by the period consists of 8 bits, represented by 0 to 255 in numerical forms. Organizations could purchase three classes of IPv4 addresses.

Class A

A Class A IPv4 address has 8 network prefix bits. For example, consider 44.0.0.1, where 44 is the network address and 0.0.1 is the host address.

Class B

A Class B IPv4 address has 16 network prefix bits. For example, consider 128.16.0.2, where 128.16 is the network address and 0.2 is the host address.

Class C

A Class C IPv4 address has 24 network prefix bits. For instance, consider 192.168.1.100, where 192.168.1 is the network address and 100 is the host address.

Classless addresses

Classless or Classless Inter-Domain Routing (CIDR) addresses use variable length subnet masking (VLSM) to alter the ratio between the network and host address bits in an IP address. A subnet mask is a set of identifiers that returns the network address's value from the IP address by turning the host address into zeroes.

A VLSM sequence allows network administrators to break down an IP address space into subnets of various sizes. Each subnet can have a flexible host count and a limited number of IP addresses. A CIDR IP address appends a suffix value stating the number of network address prefix bits to a normal IP address.

For example, 192.0.2.0/24 is an IPv4 CIDR address where the first 24 bits, or 192.0.2, is the network address.

What are the limitations of classful IP addressing that CIDR overcomes?

Before Classless Inter-Domain Routing (CIDR), IP addresses were classful and created inefficiencies. We discuss some of these shortcomings next.

Inflexible IP addressing

In a classful addressing system, each class supported a fixed number of devices:

- Class A supported 16,777,214 hosts
- Class B supported 65,534 hosts
- Class C supported 254 hosts

The classful arrangement was inefficient when allocating IP addresses and led to a waste of IP address spaces.

For example, an organization with 300 devices couldn't have used a Class C IP address, which only permitted 254 devices. So, the organization would've been forced to apply for a Class B IP address, which provided 65,534 unique host addresses. However, only 300 devices would've been connected, which would've left 65,234 unused IP address spaces.

Limitations in network design

Classful IPs limited your ability to combine networks as required. For example, these IP addresses belong to different class C networks in the classful architecture:

- 192.168.1.0
- 192.168.0.0

As a network administrator, you couldn't have combined both networks because the class C subnet mask was fixed as 255.255.255.0.

What are the benefits of CIDR?

With Classless Inter-Domain Routing (CIDR), your organization has more flexibility in assigning IP addresses and routing data between devices.

Reduce IP address wastage

CIDR provides flexibility when you determine the network and host identifier assignments on an IP address. You can use CIDR to provision the required number of IP addresses for a particular network and reduce wastage. Besides, CIDR reduces routing table entries and simplifies data packet routing.

Transmit data quickly

CIDR allows routers to organize IP addresses into multiple subnets more efficiently. A subnet is a smaller network that exists within a network. For example, all devices connected to a router are on the same subnet and have the same IP address prefix.

With CIDR, your organization can create and consolidate multiple subnets. This allows data to reach the destination address without taking unnecessary paths.

Create a Virtual Private Cloud

A virtual private cloud (VPC) is a private digital space hosted within the cloud. It allows your organization to provision workloads in an isolated and secure environment. A VPC uses CIDR IP addresses when it transfers data packets between connected devices.

Create supernets flexibly

A supernet is a group of subnets with similar network prefixes. CIDR allows flexibility in creating supernets, which isn't possible in conventional masking architecture. For example, your organization can combine IP addresses into a single network block using a notation like this:

- 192.168.1 /23
- 192.168.0 /23

This notation applies a subnet mask of 255.255.254.0 to the IP address, which returns the first 23 bits as the network address. The router needs only one routing table entry to manage data packets between devices on the subnets.

How does CIDR work?

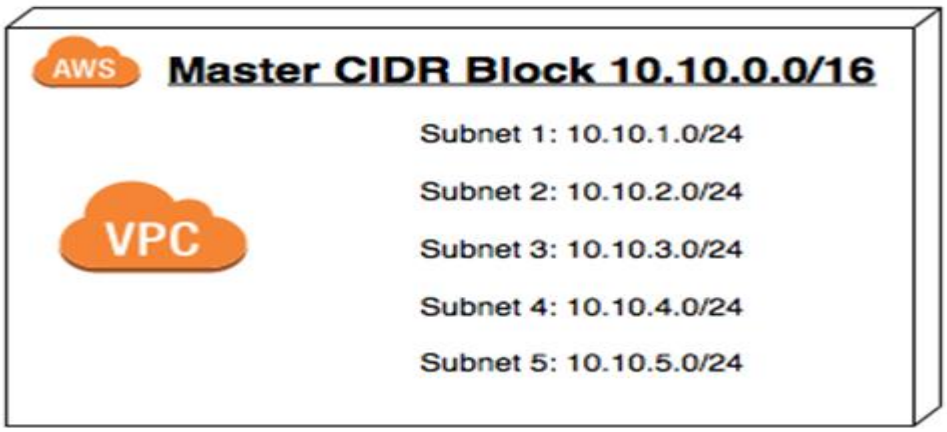
Classless Inter-Domain Routing (CIDR) allows network routers to route data packets to the respective device based on the indicated subnet. Instead of classifying the IP address based on classes, routers retrieve the network and host address as specified by the CIDR suffix.

It's important to understand CIDR blocks and CIDR notation to learn how CIDR works.

CIDR blocks

A CIDR block is a collection of IP addresses that share the same network prefix and number of bits. A large block consists of more IP addresses and a small suffix.

The Internet Assigned Numbers Authority (IANA) assigns large CIDR blocks to regional internet registries (RIR). Then, the RIR assigns smaller blocks to local internet registries (LIR), which then assign them to organizations. Meanwhile, private users apply for CIDR blocks from their internet service providers.



CIDR notation

CIDR notation represents an IP address and a suffix that indicates network identifier bits in a specified format. For example, you could express 192.168.1.0 with a 22-bit network identifier as 192.168.1.0/22.

How is CIDR used in IPv6?

IPv6 is a networking addressing system designed to replace IPv4. IPv6 uses a 128-bit unique identifier, which allows it to hold 1,028 times more IP addresses than IPv4.

An IPv6 address consists of 8 colon-separated hexadecimal values. IPv6 allows a much larger address space to accommodate the increasing number of devices that are connecting to the internet today.

Under Classless Inter-Domain Routing (CIDR), IPv6 addresses can be aggregated with prefixes of arbitrary bit length, similar to IPv4 addresses. For example, 2001:0db8:/32 is an IPv6 CIDR address, with the first 32 bits, or 2001:db8, as the network address.

What Is NAT?

NAT stands for network address translation. It's a way to map multiple private addresses inside a local network to a public IP address before transferring the information onto the internet. Organizations that want multiple devices to employ a single IP address use NAT, as do most home routers. If you're connecting from your home right now, chances are your cable modem or DSL router is already providing NAT to your home.

How Does NAT Work?

Let's say that there is a laptop connected to a home network using NAT. That network eventually connects to a router that addresses the internet. Suppose that someone uses that laptop to search for directions to their favorite restaurant. The laptop is using NAT. So, it sends this request in an IP packet to the router, which passes that request along to the internet and the search service you're using. But before your request leaves your home network, the router first changes the internal IP address from a private local IP address to a public IP address. Your router effectively translates the private address you're using to one that can be used on the internet and then back again. Now you know that your humble little cable modem or DSL router has a little, automated translator working inside of it.

If the packet keeps a private address, the receiving server won't know where to send the information back to. This is because a private IP address cannot be routed onto the internet. If your router were to try doing this, all internet routers are programmed to automatically drop private IP addresses. The nice thing is, though, that all routers sold today for home offices and small offices can readily translate back and forth between private IP address and publicly-routed IP addresses.

What are Private IP Addresses?

As the internet became more popular years ago, the organization that manages IP addresses, known as the Internet Assigned Numbers Authority (IANA) realized that they needed to do something. So, they created a network address translation scheme. This scheme is described in a document called Request for Comments (RFC) 1918. This is just one document of thousands that define how the internet works. If you want to learn about NAT, this is the document that all router manufactures must implement. No matter what type of NAT you use, you will be using RFC 1918 addresses.

If you were to try to send an RFC 1918 private IP address onto the internet, it would be much like sending a physical piece of mail with the return address of "anonymous," yet requesting return service notification. If you were to try doing that with a snail mail service, you would

never get that return service notification, because the service wouldn't be able to tell where "anonymous" even is.

NAT Types

There are three different types of NATs. People and organizations use them for different reasons, but they all still work as a NAT.

Static NAT

When the local address is converted to a public one, this NAT chooses the same one. This means there will be a consistent public IP address associated with that router or NAT device.

Dynamic NAT

Instead of choosing the same IP address every time, this NAT goes through a pool of public IP addresses. This results in the router or NAT device getting a different address each time the router translates the local address to a public address.

PAT

PAT stands for port address translation. It's a type of dynamic NAT, but it bands several local IP addresses to a singular public one. Organizations that want all their employees' activity to use a singular IP address use a PAT, often under the supervision of a network administrator.

Why Use NAT?

NAT is a straightforward process. Most routing equipment you purchase at a store will implement it automatically, or with a simple click of a mouse. Let's get a bit deeper into NAT's role in IP conservation and explain its limited role in providing security services.

IP Conservation

IP addresses identify each device connected to the internet. The existing IP version 4 (IPv4) uses 32-bit numbered IP addresses, which allows for 4 billion possible IP addresses, which seemed like more than enough when it launched in the 1970s.

However, the internet has exploded, and while not all 7 billion people on the planet access the internet regularly, those that do often have multiple connected devices: Phones, personal desktop, work laptop, tablet, TV, even refrigerators.

Therefore, the number of devices accessing the internet far surpasses the number of IP addresses available. Routing all of these devices via one connection using NAT helps to consolidate multiple private IP addresses into one public IP address. This helps to keep more public IP addresses available even while private IP addresses proliferate.

IPv6: More Addresses and Routing Efficiency

On June 6, 2012, IP version 6 (IPv6) officially launched after decades of development. IPv6 was created for many reasons. One of them was to accommodate the need for more IP addresses. This is because traditional NAT itself couldn't quite keep up with demand. IPv6 uses 128-bit numbered IP addresses, which allow for exponentially more potential IP addresses than IPv4. It will take many years before this process finishes; so until then, using NAT for IPv4 addresses will remain a common practice. More importantly, though, IPv6 does more than just provide a (much) larger IP address space. IPv6 also makes routing much more efficient. For example, IPv6 doesn't put the burden on routers to process traffic as much as IPv4 does.

NAT and Security

NAT and private IP addressing are not security services *per se*. But the use of NAT and private IP addresses is often perceived as a first step towards security. Because NAT transfers packets of data from public to private addresses, it also helps prevent outside computers from directly accessing your private device. For example, an individual on the internet would not be able to use ping or a Web browser to connect to your home computer, unless you created a very specific mapping.

Still, it is important to understand that in and of itself, NAT does not provide security services, such as firewalling, monitoring, antivirus protection, intrusion detection, application security or zero trust services. It is best to consider NAT as a service that conserves and organizes IP addresses, rather than securing information or privacy. Part of the confusion about whether or not NAT provides security services are that it implements private IP addresses. It might seem natural to assume that the term private implies security and privacy. But in almost every practical way, NAT does not provide security services. Instead, it allows you to use RFC 1918 addresses.

LECTURE 8

Basic IP Support Protocols

In this article, we will discuss the fundamental concepts and protocols that form the backbone of modern computer networks. Understanding these protocols is crucial for anyone seeking a solid foundation in networking. IP (Internet Protocol) is a fundamental building block of the Internet, enabling communication between devices across various networks. To facilitate efficient and reliable data transmission, a range of support protocols has been developed to work in conjunction with IP. These protocols provide additional functionalities and services that enhance the overall performance, security, and manageability of IP-based networks.

Throughout this study notes, we will explore key support protocols, including but not limited to Address Resolution Protocol (ARP), Internet Control Message Protocol (ICMP), Internet Group Management Protocol (IGMP), and Dynamic Host Configuration Protocol (DHCP). Each protocol plays a vital role in maintaining the smooth operation of IP networks, ensuring proper addressing, routing, error detection, and network management. Here, we are providing the complete study notes on the Basic IP Support Protocols for the preparation of the GATE, Computer Science Engineering Exam.

What are Basic IP Support Protocols?

Basic IP Support Protocols are essential components of the modern Internet infrastructure that facilitate efficient communication and data transfer across networks. These protocols play a crucial role in establishing and maintaining connections between devices, enabling the

seamless exchange of information. Understanding these fundamental protocols is essential for network administrators, engineers, and anyone involved in managing and troubleshooting network connectivity.

IP, or Internet Protocol, serves as the backbone of the Internet, responsible for addressing and routing packets of data between devices. However, the successful transmission of data relies on additional support protocols that address various aspects of network communication. Basic IP Support Protocols encompass a range of protocols that assist in tasks such as host configuration, error detection and correction, and network management.

Address Resolution Protocol (ARP)

Address Resolution Protocol (ARP) is one of the major protocols of the TCP/IP protocol suite. The main goal of Address Resolution Protocol (ARP) is to map an IPv4 address (32-bit Logical Address) to its corresponding MAC Address (48-bit Physical Address). Network Applications use IPv4 Addresses to communicate with other devices at the Application Layer. The data link layer deal with the MAC address (48-bit Physical Address) for addressing, and this address (physical address) is configured into the network card of the system permanently.

Note: ARP determines the hardware address/physical address, also called the Media Access Control (MAC) address, of a host from its well-known IP address.

The important terms associated with ARP are :

1. **ARP Cache:** After resolving the MAC address, the ARP sends it to the source where it stores in a table for future reference. The subsequent communications can use the MAC address from the table
2. **ARP Cache Timeout:** It indicates the time for which the MAC address in the ARP cache can reside
3. **ARP request:** This is nothing but broadcasting a packet over the network to validate whether we came across the destination MAC address or not.

ARP request packet contains:

1. The physical address of the sender.
2. The IP address of the sender.
3. The physical address of the receiver is zero seconds.
4. The IP address of the receiver.

Note, that the ARP packet is encapsulated directly into the data link frame.

1. **ARP response/reply:** It is the MAC address response that the source receives from the destination which aids in further communication of the data.

Cases When ARP is Used

- **Case-1:** The sender is a host and wants to send a packet to another host on the same network.
    - Use ARP to find another host's physical address
- **Case-2:** The sender is a host and wants to send a packet to another host on another network.
    - The sender looks at its routing table.
    - Find the IP address of the next hop (router) for this destination.
    - Use ARP to find the router's physical address
- **Case-3:** the sender is a router and received a datagram destined for a host on another network.
    - The router checks its routing table.
    - Find the IP address of the next router.
    - Use ARP to find the next router's physical address.
- **Case-4:** The sender is a router that has received a datagram destined for a host in the same network.
    - Use ARP to find this host's physical address.

NOTE: An ARP request is broadcast, and an ARP response is a Unicast.

What is Dynamic Host Configuration Protocol?

In simple words, DHCP (Dynamic Host Configuration Protocol) can be defined as a set of rules that is responsible for the dynamic configuration and management of hosts in a network. It dynamically assigns an IP address and other network managing parameters to the system so that it can communicate with other devices (IP networks).

- Dynamic – Automatically
- Host – Any computer that is connected to the network
- Configuration – To configure a host means to provide network information(IP address, subnet mask, Gateway address) to a host
- Protocol – Set of rules

Basically, the main role of a DHCP server is to configure a host in a network dynamically.

It is an application layer protocol that provides:

1. Subnet Mask (Option 1 – e.g., 255.255.255.0)
2. Router Address (Option 3 – e.g., 192.168.1.1)

3. DNS Address (Option 6 – e.g., 8.8.8.8)
4. Vendor Class Identifier (Option 43 – e.g., 'unifi' = 192.168.1.9 ##where unifi = controller)

DHCP is based on a client-server model and based on discovery, offer, request, and ACK.

Here is how the DHCP process works when you go online

1. You go on your computer to connect to the Internet.
2. The network requests an IP address (this is actually referred to as a DHCP discover message).
3. On behalf of your computer's request, the DHCP server allocates (leases) to your computer an IP address. This is referred to as the DHCP offer message.
4. Your computer (remember—you're the DHCP client) takes the first IP address offer that comes along. It then responds with a DHCP request message that verifies the IP address that's been offered and accepted.
5. DHCP then updates the appropriate network servers with the IP address and other configuration information for your computer.
6. Your computer (or whatever network device you're using) accepts the IP address for the lease term.

DHCP discover message

This is the first message in the communication process between any client and server. Discovr message is provoked by the client host to discover any available DHCP server/servers in the network. So this message is broadcasted in the network and received by each device of that network in order to determine the DHCP server.

DHCP offer message

This message is sent as a response from the DHCP server to the requesting host in this message specifying the unleased IP address with other TCP configuration information. DHCP offer message is broadcasted by the server.

DHCP request message

When a requesting client gets an offer message, it again acknowledges it by broadcasting a DHCP request message. The client will generate a gratuitous ARP to find whether there is any other host present in the network with the same IP address. If the client gets no reply from the other host, so t means there is no host with the same TCP configuration in the network, and then the message is broadcasted to the server acknowledging the acceptance of the IP address, and a Client ID is also added in DHCP request message.

Note DHCP request message is broadcasted after the ARP request is broadcasted by the host to check whether any other host is not using offered IP address. If there is no reply from any other host, then the client host broadcast the DHCP request message to the server acknowledging the acceptance of the IP address with other TCP/IP Configuration.

DHCP acknowledgment message

A DHCP acknowledgment message is sent in response to the request message received, the DHCP server will make an entry specifying the client ID and then bind the offered IP address with the lease time. Finally, the client will have the IP address provided by the server.

DHCP negative acknowledgment message

Whenever a DHCP server receives a request for an IP address that is invalid so it sends a DHCP Nak (Negative acknowledgment) message to the client. For example -when the server has no unused IP address or the pool is empty, then a DHCP Nak message is sent to the client by the server.

Advantages – The advantages of using DHCP include

- Centralized management of IP addresses
- Ease of adding new clients to a network
- Reuse of IP addresses reduces the total number of IP addresses that are required
- Simple reconfiguration of the IP address space on the DHCP server without needing to reconfigure each client

Internet Control Message Protocol (ICMP)

ICMP (Internet Control Message Protocol) is an Internet layer protocol used by network devices to diagnose network communication issues. Mainly ICMP is used to determine whether data is reaching desired correct destination in a timely manner or not. Generally, ICMP protocol is used on network devices, such as routers, gateway, etc.

What Is ICMP Used For?

IP (Internet Protocol) doesn't have any inbuilt mechanism for detecting and sending error and control messages. So, IP depends on Internet Control Message Protocol (ICMP) to provide error control. It is used for error reporting and management-related queries. ICMP is a supporting protocol that is mostly used by networking devices such as routers for sending error messages and operations information.

When two devices are connected over the Internet, the ICMP generates errors to share with the sending device so that any of the data didn't get to its intended destination.

Source Quench Message

A source quench message is a request to the host(destination) to decrease the traffic rate for message sending. Or in simple words, we can say that when receiving host detects that the traffic rate (rate of sending packets) is too fast, then it sends the source quench message to the source to slow down the message sending speed (rate) so that no packet can be lost.

- ICMP will take the source IP from the discarded packet and informs to source by sending the source quench message. The source will reduce the speed of transmission so that router will be free from congestion.
- When the congestion router is far away from the source the ICMP will send a hop-by-hop source quench message so that every router will reduce the speed of transmission.

Parameter Problem

Whenever the router receives data packets then the calculated header checksum should always be equal to the received header checksum then only the packet is accepted by the router otherwise rejected.

- If there is a mismatch packet will be dropped by the router. ICMP will take the source IP from the discarded packet and informs to source by sending a parameter problem message.

Time Exceeded Message

- When some fragments are lost in a network then the holding fragment by the router will be dropped then ICMP will take the source IP from the discarded packet and informs the source, of discarded datagram due to the time to live field reaches to zero, by sending the time exceeded message.

Destination Un-reachable

- The destination is unreachable and is generated by the host or its inbound gateway to inform the client that the destination is unreachable for some reason.
- There is no necessary condition that only the router gives the ICMP error message some time destination host sends an ICMP error message when any type of failure (link failure, hardware failure, port failure, etc.) happens in the network.

Redirection Message

- Redirect requests data packets be sent on an alternate route. The message informs to a host to update its routing information (to send packets on an alternate route).
- If the host tries to send data through a router R_1 and R_1 sends data on a router R_2 and there is a direct way from the host to R_2. Then R_1 will send a redirect message to inform the host that there is a best way to the destination directly through

R_2 available. The host then sends data packets for the destination directly to R_2. The router R_2 will send the original datagram to the intended destination.

- But if datagram contains the routing information of the packet then this message will not be sent even if another better route is available. As redirected messages should only be sent by the gateways and not by the Internet hosts.
- Whenever a packet is forwarded in the wrong direction later it is redirected in a current direction and then ICMP will send the re-directed message.

Network Address Translation (NAT)

- To access the Internet, one public IP address is needed, but we can use a private IP address in our private network. The idea of NAT is to allow multiple devices to access the Internet through a single public address. To achieve this, the translation of a private IP address to a public IP address is required.
- Network Address Translation (NAT)is a process in which one or more local IP address is translated into one or more Global IP address and vice versa in order to provide Internet access to the local hosts.
- Also, it does the translation of port numbers i.e. masks the port number of the host with another port number, in the packet that will be routed to the destination. It then makes the corresponding entries of IP address and port number in the NAT table. NAT generally operates on a router or firewall.

Network Address Translation (NAT) working

Mostly, Network Address Translation is configured with the border router i.e. the router which has one interface in the local network and one interface in the global network. When a packet travels outside the local (inside) network, then NAT transforms that local (private) IP address into a global (public) IP address. When a packet enters into the local network, the global (public) IP address is changed into a local (private) IP address.

If NAT addresses got finished, i.e., no more address is left in the configuring pool then the packets will be dropped and an Internet Control Message Protocol (ICMP) host unreachable packet will be sent to the destination.

Advantages of NAT

- NAT conserves legally registered IP addresses.
- It provides privacy as the device's IP address, sending and receiving the traffic, will be hidden.
- Eliminates address renumbering when a network evolves.

The disadvantage of NAT

- Translation results in switching path delays.
- Certain applications will not function while NAT is enabled.
- Complicates tunneling protocols such as IPsec.
- Also, the router being a network layer device, should not tamper with port numbers(transport layer) but it has to do so because of NAT.